Advanced micro lab

Lab report 2

Shams tarek

Aya elbakri

Hanna mounir

ALU C code

```c
#include <stdio.h>
#include <stdint.h>

#define add 0b000
#define sub 0b001
#define mult 0b010
#define and 0b011
#define or 0b100
#define xor 0b101
#define SHL 0b110
#define SHR 0b111

__int32_t ALU(__int32_t x, __int32_t y, __int32_t
select);

int main() {
   __int32_t a = 12;
   __int32_t b = 3;

   printf("Add: %d\n", ALU(a, b, add));
   printf("Sub: %d\n", ALU(a, b, sub));
   printf("Mult: %d\n", ALU(a, b, mult));
   printf("AND: %d\n", ALU(a, b, and));
   printf("OR : %d\n", ALU(a, b, or));
   printf("XOR: %d\n", ALU(a, b, xor));
   printf("SHL: %d\n", ALU(a, b, SHL));
   printf("SHR: %d\n", ALU(a, b, SHR));

   return 0;
}
```

```c
__int32_t ALU(__int32_t x, __int32_t y, __int32_t
select){

   __int32_t result = 0;

   if (select == add){
      result = x + y;
   }
   else if (select == sub){
      result = x - y;
   }
   else if (select == mult){
      result = x * y;
   }
   else if (select == and){
      result = x & y;
   }
   else if (select == or){
      result = x | y;
   }
   else if (select == xor){
      result = x ^ y;
   }
   else if (select == SHL){
      result = x << y;
   }
   else if (select == SHR){
      result = (__uint32_t)x >> y;
   }
   return result;    }
```

Complied code

```
asiclab@ASICLab:~/bambu$ ./ALU
Add: 15
Sub: 9
Mult: 36
AND: 0
OR : 15
XOR: 15
SHL: 96
SHR: 1
asiclab@ASICLab:~/bambu$
```

ALU_TEST BENCH

```
`timescale 1ns / 1ps

module add_tb;

reg clock;
  reg reset;
  reg start_port;
  reg [31:0] x;
  reg [31:0] y;
  reg [31:0] select;
  wire done_port;
  wire [31:0] return_port;

  ALU uut (
    .clock(clock),
    .reset(reset),
    .start_port(start_port),
    .x(x),
    .y(y),
    .select(select),
    .done_port(done_port),
    .return_port(return_port)
  );

  always #5 clock = ~clock;

  task run_test;
    input [31:0] a;
    input [31:0] b;
    input [31:0] sel;
    begin
      x = a;
      y = b;
      select = sel;
      @(posedge clock);
      start_port = 1;

      @(posedge clock);
      start_port = 0;
      wait(done_port);
      @(posedge clock);

      $display("Time=%0t | select=%0d | x=%0d |
y=%0d | result=%0d",
              $time, sel, a, b, return_port);

      repeat(2) @(posedge clock);
    end
  endtask
  initial begin
    $dumpfile("simulation1.vcd");
    $dumpvars(0, add_tb);

    clock = 0;
    reset = 0;
    start_port = 0;
    x = 0;
    y = 0;
    select = 0;

    #20;
    reset = 1;

    run_test(12, 3, 3'b000);   // ADD
    run_test(12, 3, 3'b001);   // SUB
    run_test(12,  3, 3'b010);   // MUL
    run_test(12, 3, 3'b011);  // AND
    run_test(12, 3, 3'b100);  // OR
    run_test(12, 3, 3'b101);  // XOR
    run_test(12,  3, 3'b110);   // LEFT SHIFT
    run_test(12, 3, 3'b111);   // RIGHT SHIFT

    #50;
    $finish;
  end

endmodule
```

Bambu

Timing report

```
Scheduling Information of function ALU:
  Number of control steps: 5
  Minimum slack: 1.0553999990000034
  Estimated max frequency (MHz): 111.79929788791016
Time to perform scheduling: 0.00 seconds
```

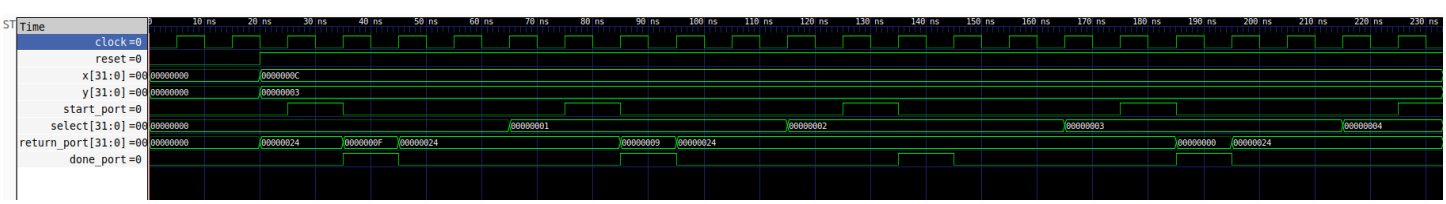
Utilization

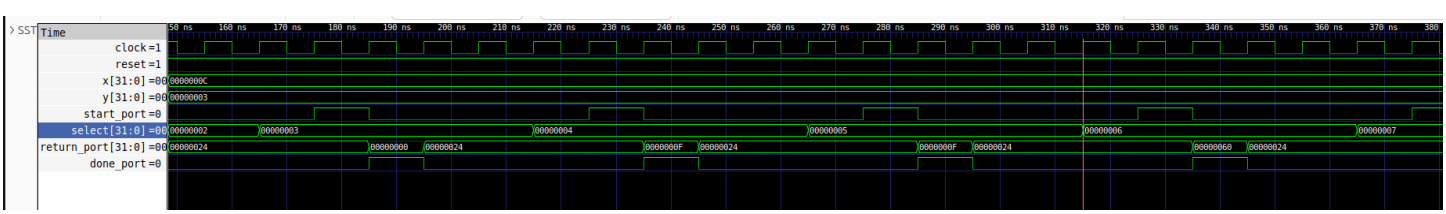
```
Module binding information for function ALU:
  Number of modules instantiated: 38
  Number of possible conflicts for possible false paths introduced by resource sharing: 0
  Estimated resources area (no Muxes and address logic): 773
  Estimated area of MUX21: 0
  Total estimated area: 773
  Estimated number of DSPs: 3
  Slack computed in 0.00 seconds
  False-loop computation completed in 0.00 seconds
  Weight computation completed in 0.00 seconds
  Clique covering computation completed in 0.00 seconds
Time to perform module binding: 0.00 seconds
```

SIM TB

```
asiclab@ASICLab:~/shams_sem10/ALU$ iverilog -o sim.vvp ALU.v ALU_TB.v
asiclab@ASICLab:~/shams_sem10/ALU$ vvp sim.vvp
VCD info: dumpfile simulation.vcd opened for output.
Time=45000 | select=0 | x=12 | y=3 | result=15
Time=95000 | select=1 | x=12 | y=3 | result=9
Time=145000 | select=2 | x=12 | y=3 | result=36
Time=195000 | select=3 | x=12 | y=3 | result=0
Time=245000 | select=4 | x=12 | y=3 | result=15
Time=295000 | select=5 | x=12 | y=3 | result=15
Time=345000 | select=6 | x=12 | y=3 | result=96
Time=395000 | select=7 | x=12 | y=3 | result=1
ALU_TB.v:90: $finish called at 465000 (1ps)
```

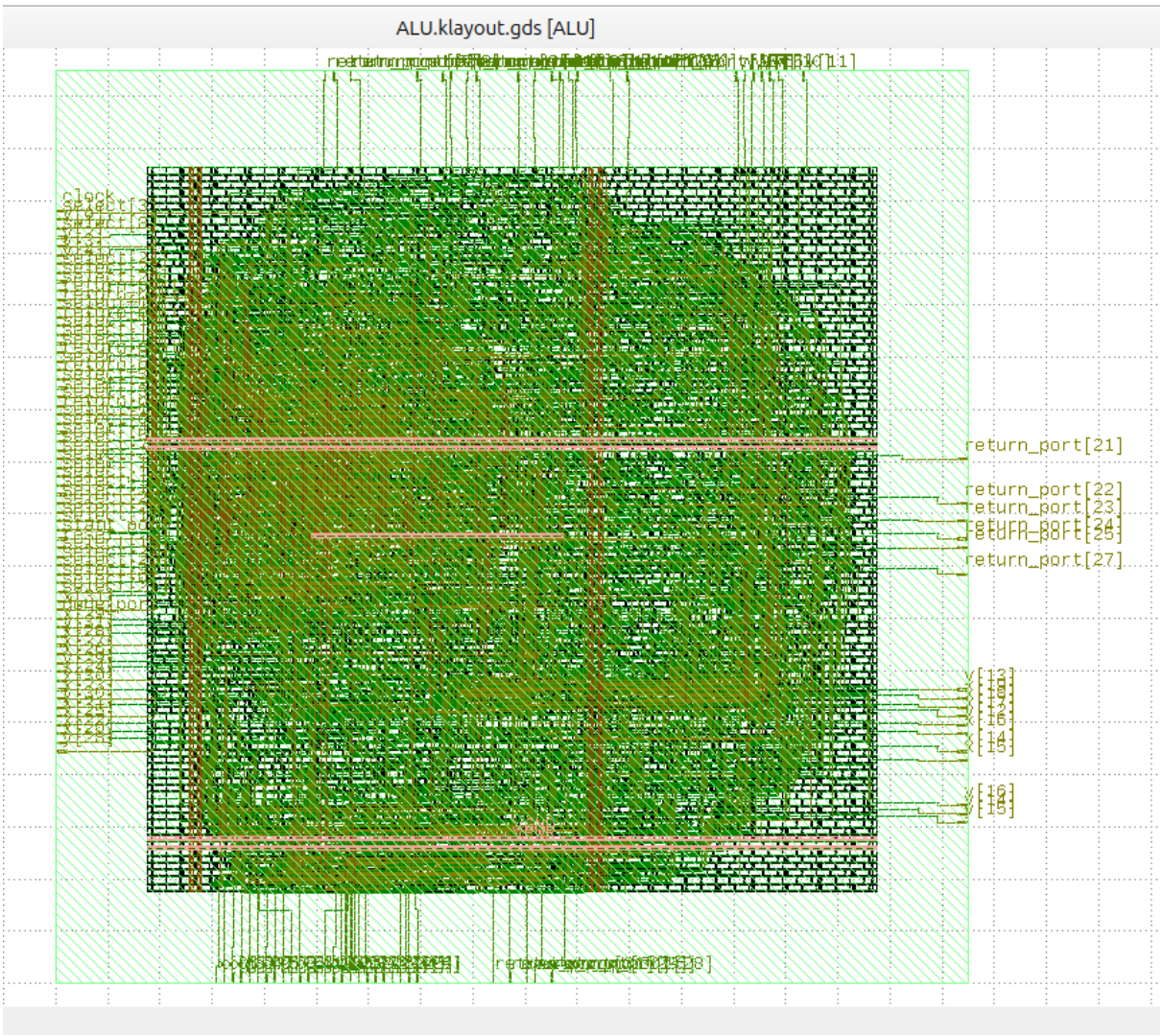
Waveform





Librelane

ALU GDS



Area report

Config.json

{“DESIGN_NAME": "ALU",

"VERILOG_FILES": ["dir::ALU.v"],

"CLOCK_PERIOD": 5,

"CLOCK_PORT": "clock",

"DIE_AREA": [0,0,350,350],

"CORE_AREA": [35,35,315,315],

"FP_SIZING": "absolute”}

Power report

```
============================================================================
 report_power
============================================================================
======================= max_tt_025C_1v80 Corner ===================================

Group                    Internal    Switching      Leakage        Total
                            Power        Power        Power        Power (Watts)
----------------------------------------------------------------------------
Sequential           2.907828e-04 2.349789e-06 2.915856e-10 2.931328e-04   1.1%
Combinational        1.158764e-02 1.408746e-02 1.515996e-08 2.567512e-02  97.3%
Clock                2.355714e-04 1.884690e-04 7.736258e-09 4.240481e-04   1.6%
Macro                0.000000e+00 0.000000e+00 0.000000e+00 0.000000e+00   0.0%
Pad                  0.000000e+00 0.000000e+00 0.000000e+00 0.000000e+00   0.0%
----------------------------------------------------------------------------
Total                1.211400e-02 1.427828e-02 2.318778e-08 2.639230e-02 100.0%
                            45.9%        54.1%         0.0%
```

Timing report

```
grep slack max.rpt
                                  -4.838003   slack (VIOLATED)
                                  -4.758920   slack (VIOLATED)
                                  -4.745209   slack (VIOLATED)
                                  -4.645964   slack (VIOLATED)
                                  -4.569246   slack (VIOLATED)
                                  -4.549970   slack (VIOLATED)
                                  -4.446833   slack (VIOLATED)
                                  -4.434013   slack (VIOLATED)
                                  -4.413581   slack (VIOLATED)
                                  -4.406947   slack (VIOLATED)
```

```
grep "data arrival time" max.rpt
                                   8.588002   data arrival time
                                  -8.588002   data arrival time
                                   8.508920   data arrival time
                                  -8.508920   data arrival time
                                   8.495210   data arrival time
                                  -8.495210   data arrival time
                                   8.395965   data arrival time
```

Total Cell Area	Core Area	Critical Path Delay	Max Operating Frequency	Power	Setup violation	Worst Slack
0, 0, 350, 350	35,35,315,315	8.588 ns	101.6 MHz	26.4mW	-4.838ns	0.369ns

Critical path description

Startpoint: x[12] (input port clocked by clock)
Endpoint: return_port[31] (output port clocked by clock)
Path Group: clock
Path Type: max